

Project Documentation

1. Data Cleaning

The following cleaning and validation steps have been completed:

- Replaced all **null values** in the children column with **0**.
 - Changed the data type of the agent and company columns from numeric to **Text**, since they represent identifiers rather than numerical values.
 - Identified an outlier in the adr (Average Daily Rate) column where a booking for **Daniel Walter** had a value of **5400**.
 - Verified the value with the data manager and corrected it to **540**.
 - Performed a complete data quality check by:
 - Checking for remaining null values.
 - Looking for inconsistencies and invalid values.
 - Verifying the data types of all columns.
 - Confirming the dataset is clean and ready for analysis.
 - **Agent ID Correction:** A large number of bookings had missing values in the agent column. After consulting the data manager, it was confirmed that these missing values should correspond to **Agent ID 22**. The missing values were replaced with 22 to ensure data completeness and accurate analysis.
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2. Data Modeling

To improve performance and follow dimensional modeling best practices, the original table was normalized into a **Star Schema**.

Fact Table

- HotelBooking

Dimension Tables

- Date
- Country
- Hotels
- Meal
- MarketSegment
- DistributionChannel
- CustomerType
- DepositType

- Customers (contains customer identification information)

Relationships were configured as **One-to-Many**, with all dimensions connected directly to the **HotelBooking** fact table.

3. Country Dimension

- Imported a separate **Country Code** table.
 - Renamed the original country column to country_code.
 - Related the fact table to the Country dimension using the country code.
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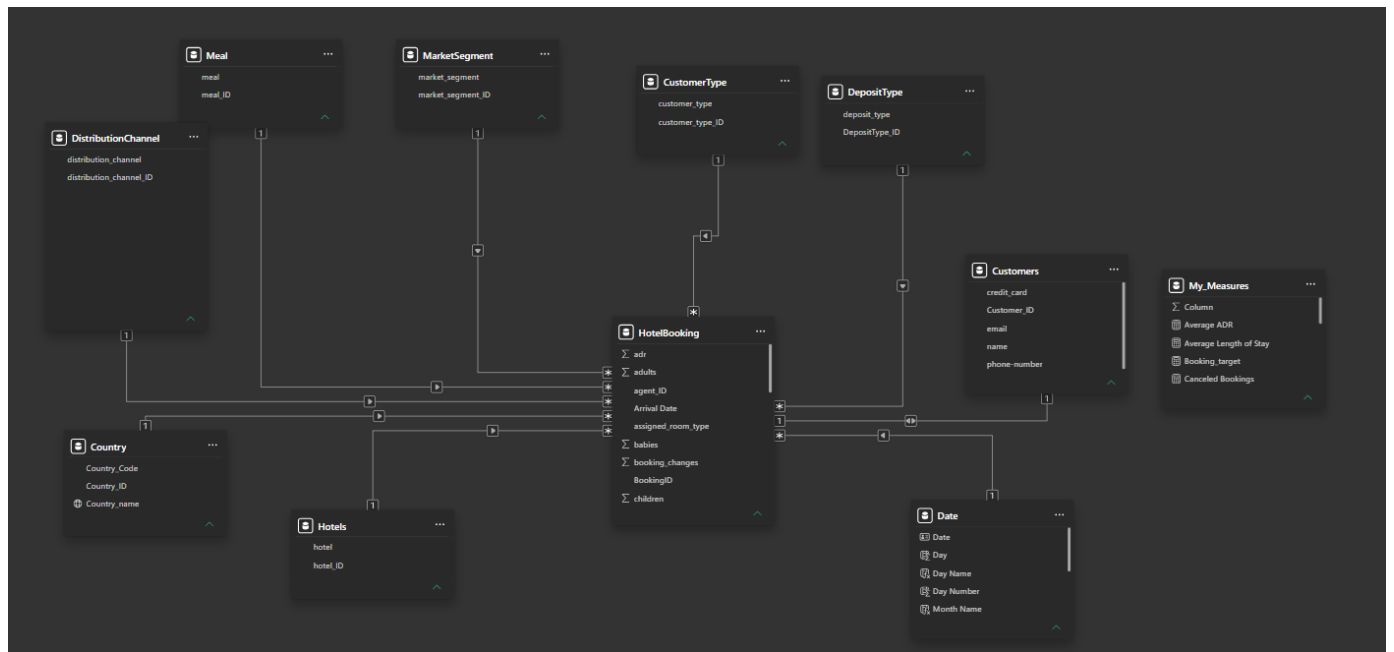
4. Date Dimension

A dedicated **Date Dimension** was created to support time-based analysis.

It includes attributes such as:

- Date
- Year
- Quarter
- Month
- Month Number
- Week Number
- Day
- Day Name
- Weekend / Weekday

The Date table is linked to the fact table through the **Arrival Date**.



Measures

Average ADR =

AVERAGE(HotelBooking[adr])

Average Length of Stay =

```
DIVIDE(  
    [Total Nights],  
    [Total Bookings],  
    0  
)
```

Canceled Bookings =

```
CALCULATE(  
    COUNT(HotelBooking[BookingID]),  
    HotelBooking[is_canceled] = 1  
)
```

Cancellation Rate =

```
DIVIDE(  
    [Canceled Bookings],  
    [Total Bookings],  
    0  
)
```

Changed Bookings = CALCULATE([Total Bookings], HotelBooking[booking_changes] > 0)

Checked In = CALCULATE([Total Bookings], HotelBooking[reservation_status] = "Check-Out")

Confirmed Bookings = CALCULATE([Total Bookings], HotelBooking[reservation_status] <> "Canceled")

Repeat Guest % =

```
DIVIDE(  
    [Repeat Guests],
```

```
[Total Bookings],  
0  
)
```

```
Repeat Guests =  
CALCULATE(  
    COUNT(HotelBooking[BookingID]),  
    HotelBooking[is_repeated_guest] = 1  
)
```

```
Total Adults = SUM(HotelBooking[adults])
```

```
Total Babies = SUM(HotelBooking[babies])
```

```
Total Bookings =  
COUNT(HotelBooking[BookingID])
```

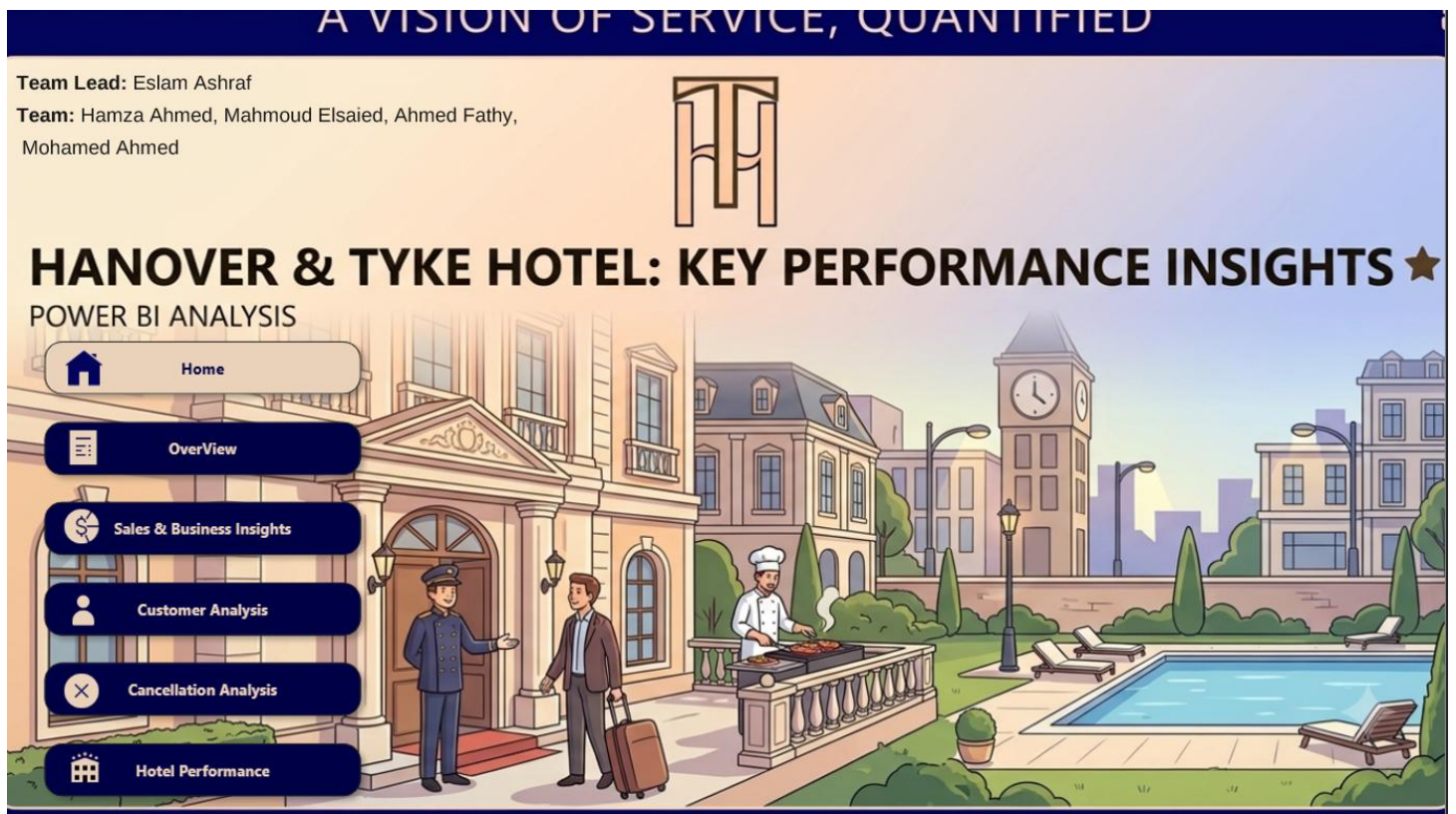
```
Total Children = SUM(HotelBooking[children])
```

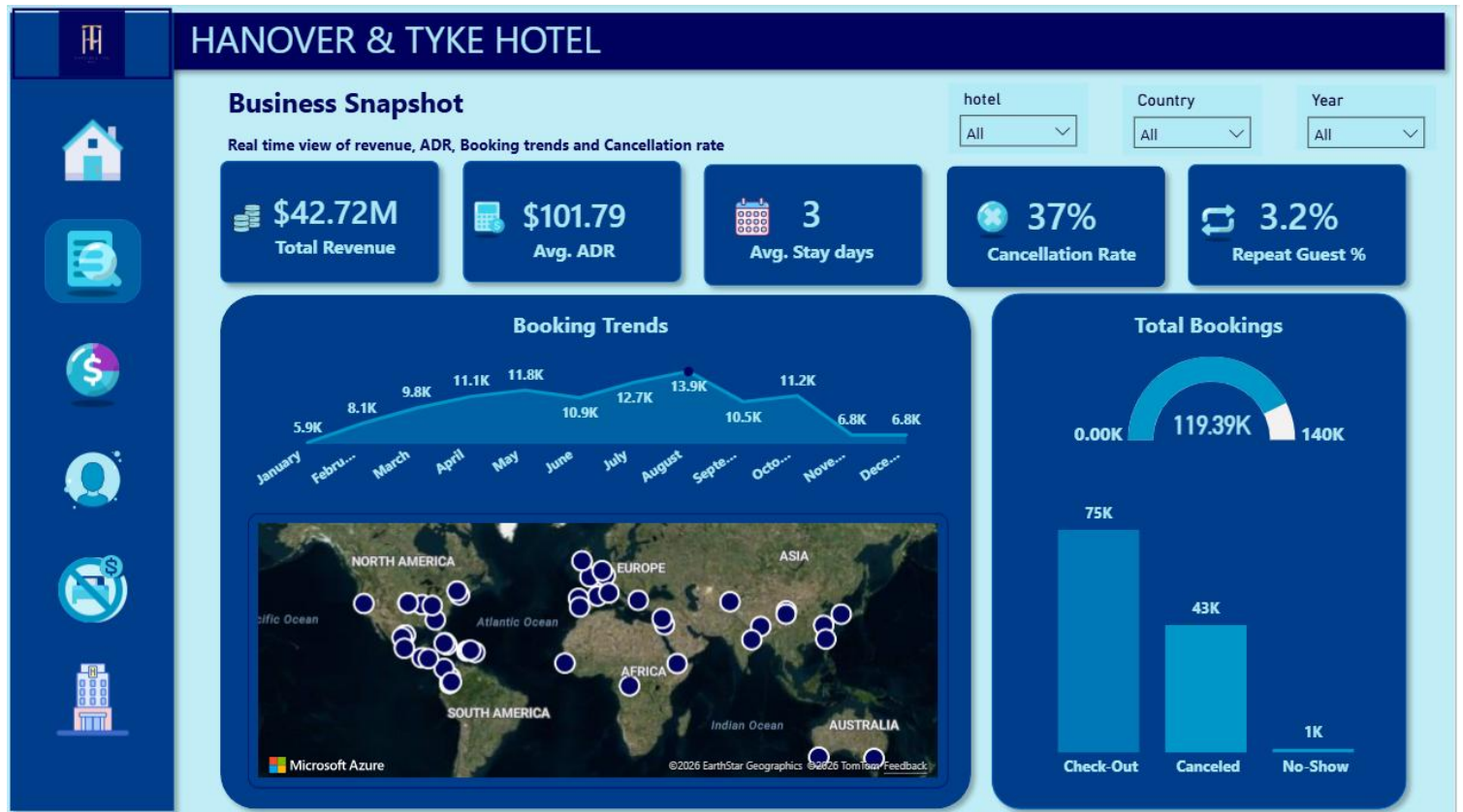
```
Total Nights =  
SUM(HotelBooking[stays_in_weekend_nights]) +  
SUM(HotelBooking[stays_in_week_nights])
```

```
Total Revenue =  
SUMX(  
    HotelBooking,  
    HotelBooking[adr] *  
    (  
        HotelBooking[stays_in_weekend_nights] +  
        HotelBooking[stays_in_week_nights]  
    )  
)
```

)

Total Special Requests = SUM(HotelBooking[total_of_special_requests])





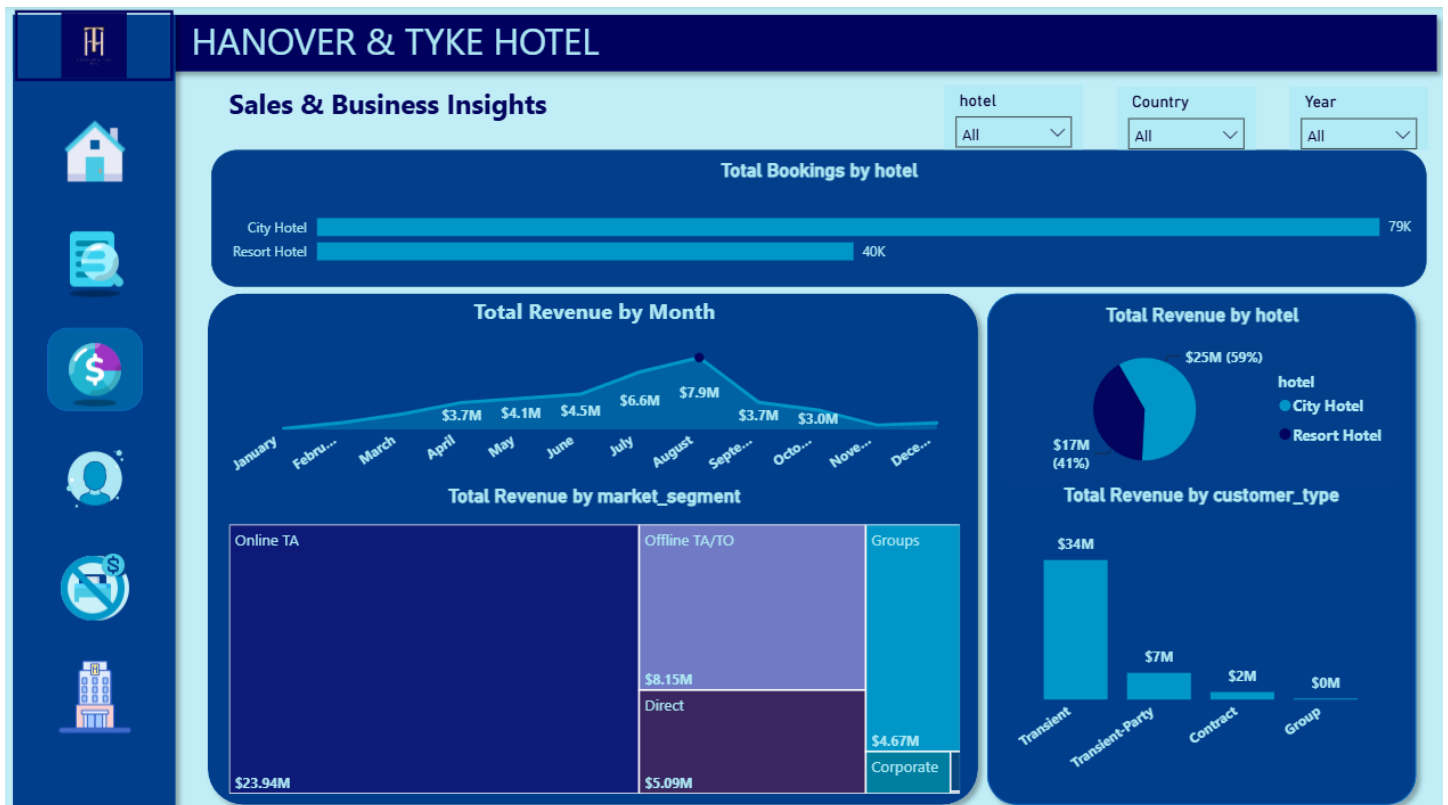
1. Business Snapshot

Key Insights:

- **Revenue & Volume:** The hotel generated \$42.72M in total revenue with an Average Daily Rate (ADR) of \$101.79.
- **Leakage:** The cancellation rate sits at an alarming 37% (43K canceled bookings vs. 75K check-outs).
- **Retention:** Only 3.2% of guests are repeat customers, indicating that the business relies heavily on acquiring new customers rather than retaining existing ones.
- **Seasonality:** Bookings and revenue follow a strict summer seasonality curve, peaking in August (13.9K bookings) and dropping significantly in November and December (6.8K bookings).

Strategic Recommendations:

- **Implement a Loyalty Program:** With retention at just 3.2%, creating a tiered rewards program or offering direct-booking return discounts is essential to increase the lifetime value of guests and reduce acquisition costs.
- **Smooth Seasonal Dips:** Launch targeted winter campaigns, corporate retreat packages, or holiday specials to increase occupancy during the November–January slump.



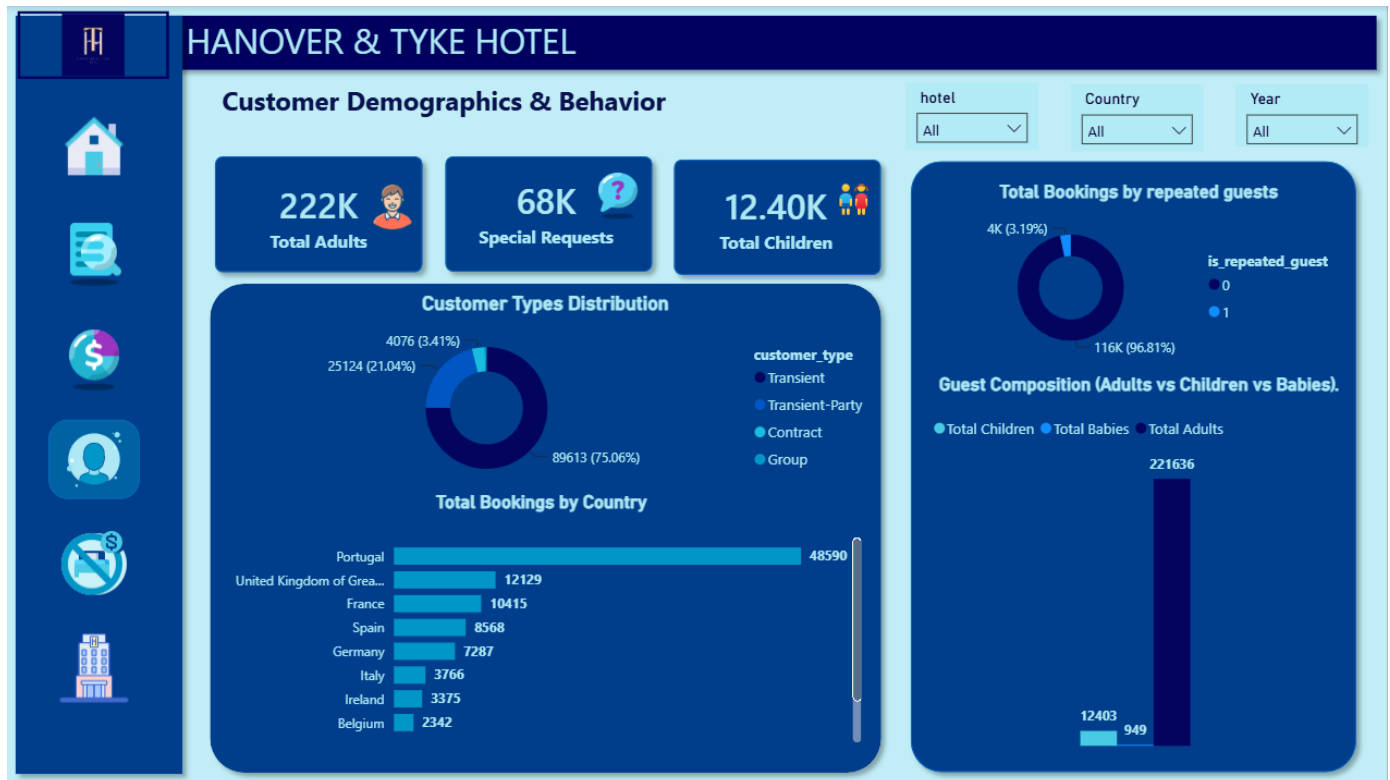
2. Sales & Business Insights

Key Insights:

- **Hotel Comparison:** The City Hotel handles the bulk of the business (79K bookings, \$25M revenue) compared to the Resort Hotel (40K bookings, \$17M revenue).
- **Distribution Channels:** Online Travel Agencies (OTAs) are dominating revenue generation (\$23.94M), while Direct bookings lag significantly behind at only \$5.09M.
- **Revenue by Month:** Revenue closely tracks the booking trends, cresting in August (\$7.9M) and bottoming out in the late fall/winter.

Strategic Recommendations:

- **Shift Share to Direct Bookings:** OTAs charge high commissions (often 15-25%). Launch a "Book Direct & Save" initiative offering perks like free breakfast, late checkout, or a 5% discount to move traffic away from OTAs and improve profit margins.
- **Leverage the City Hotel:** Since the City Hotel drives 59% of revenue, consider optimizing its pricing dynamically. During peak corporate seasons, ADR can likely be pushed higher.



2. Customer Demographics & Behavior

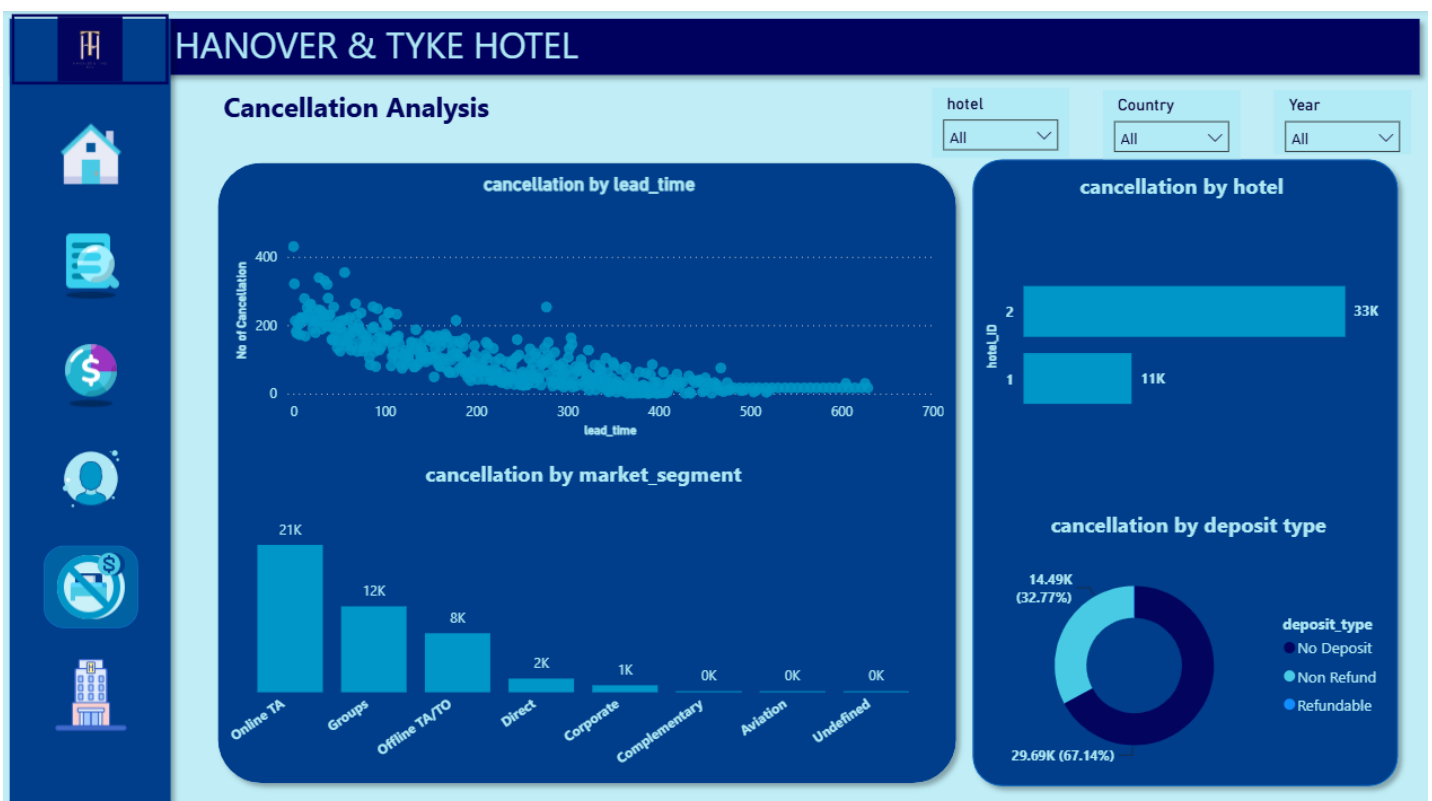
Key Insights:

- **Geographic Concentration:** Portugal is overwhelmingly the top market (48,590 bookings), driving roughly four times the volume of the next highest country (UK at 12,129).
- **Guest Composition:** The clientele is almost entirely adults (222K) with very few children (12.4K) or babies (949).
- **Customer Type:** "Transient" (individual, non-group travelers) dominates at 75.06%. "Transient-Party" follows at 21.04%, with virtually no Group or Contract business.

Strategic Recommendations:

- **Targeted Marketing:** Double down on localized marketing in Portugal, the UK, and France. Ensure the website and booking engines are natively translated and optimized for these regions.

- **Refine the Hotel's Positioning:** Since the demographic is heavily adult and transient, shift amenities and marketing to appeal to couples, solo travelers, and business professionals rather than investing heavily in family or children's entertainment.



4. Cancellation Analysis

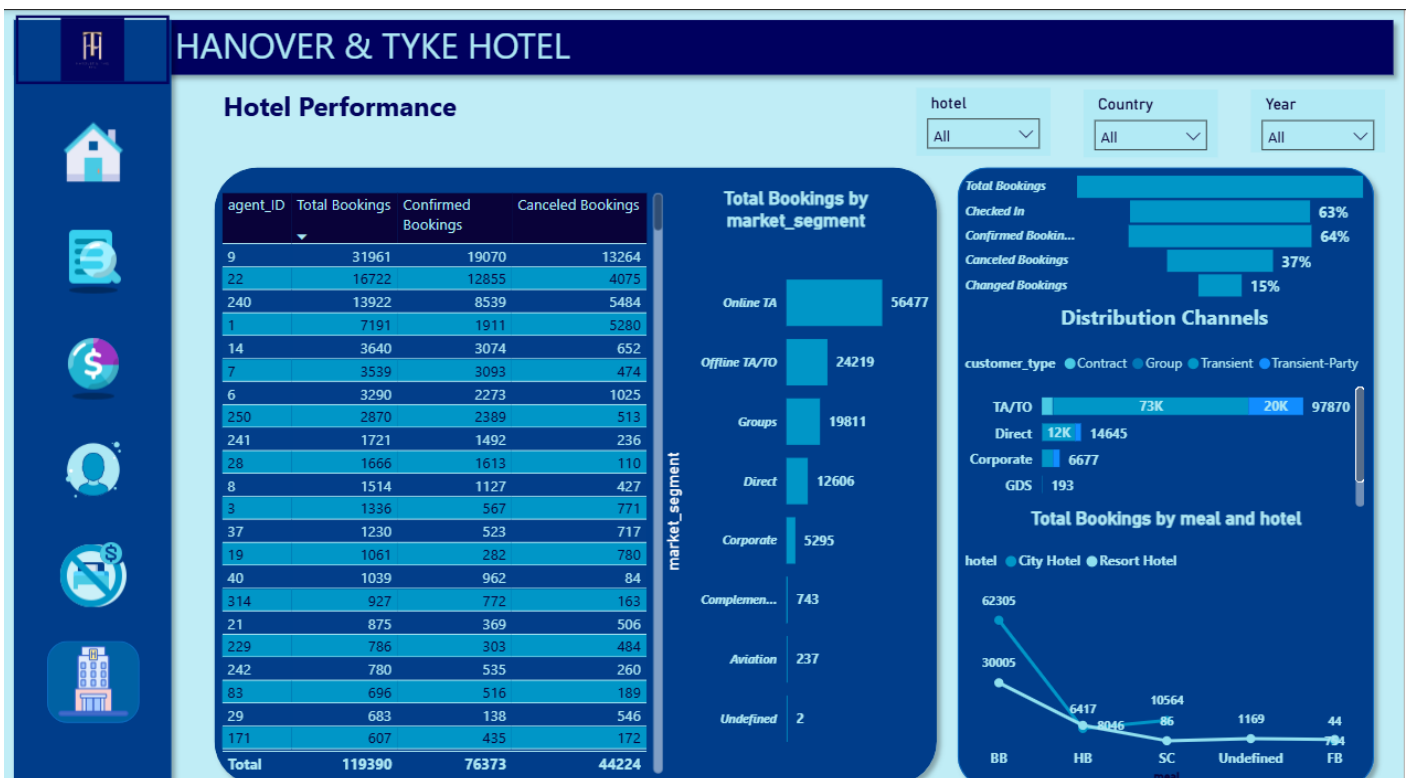
Key Insights:

- **Deposit Types:** 67.14% of cancellations come from "No Deposit" bookings. Surprisingly, 32.77% of cancellations come from "Non Refund" bookings.
- **Lead Time:** The scatter plot shows a massive cluster of cancellations occurring when the lead time is under 200 days.
- **Segment Risk:** Online TAs account for the highest volume of cancellations (21K), followed by Groups (12K).

- **Property Discrepancy:** Hotel 2 experiences three times the cancellation volume (33K) as Hotel 1 (11K).

Strategic Recommendations:

- **Overhaul Deposit Policies:** The "No Deposit" policy is creating ghost bookings. Require a first-night deposit for OTA bookings to guarantee intent.
- **Investigate "Non-Refund" Cancellations:** Clarify if the hotel is retaining the revenue for the 14.49K "Non Refund" cancellations. If OTAs are forcing refunds through loopholes, contracts need to be renegotiated.
- **Implement Overbooking Strategies:** Since OTA and Group cancellations are highly predictable, utilize data-driven overbooking limits for these specific segments to ensure maximum occupancy.



5. Hotel Performance

Key Insights:

- **Agent Performance:** Agent ID 9 is bringing in massive volume (31,961 bookings) but has an exceptionally high cancellation count (13,264).
- **Meal Plans:** Bed & Breakfast (BB) is the undisputed top meal plan (62K+ at City, 30K+ at Resort). Half Board (HB) and Self Catering (SC) are rarely utilized.
- **Funnel Drop-off:** The distribution funnel shows a 37% drop-off due to canceled bookings, leaving check-ins at roughly 63% of total top-of-funnel activity.

Strategic Recommendations:

- **Agent Accountability:** Audit the quality of traffic coming from Agent 9 and Agent 240. If they are incentivized purely on booking volume, they may be pushing low-intent traffic. Introduce commission structures that reward *completed stays* rather than raw bookings.
- **Upsell Meal Plans:** Since almost everyone chooses BB, create automated pre-arrival emails offering a discounted upgrade to Half Board (HB). This captures extra food and beverage revenue before the guest even arrives on the property.